

Multiple Choice

1. **Answer:** C

Options: A) $P = mv(t + g)$; B) $P = m(v + gt)^2$; C) $P = mgvt + mv^2 = mv(v + gt)$; D) $P = mv^2 - mgvt$; E) $P = mv + gt$

Note: Correct option: C - $P = mgvt + mv^2 = mv(v + gt)$

2. **Answer:** A

Options: A) 72,000 calories; B) 80,000 calories; C) 36,000 calories; D) 65,000 calories; E) 48,000 calories

Note: Correct option: A - 72,000 calories

3. **Answer:** D

Options: A) Deimos and Triton; B) Phobos and Demos; C) Triton and Deimos; D) Phobos and Deimos; E) Tritos and Desmos

Note: Correct option: D - Phobos and Deimos

4. **Answer:** A

Options: A) 2.65 e-07; B) 7.10 e-07; C) 2.30 e-07; D) 8.45 e-07; E) 5.67 e-07

Note: Correct option: A - 2.65 e-07

5. **Answer:** A

Options: A) 5 foot-candles; B) 1 foot-candle; C) 50 foot-candles; D) 15 foot-candles; E) 20 foot-candles

Note: Correct option: A - 5 foot-candles

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '7.5 lbs. of force'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '(5 / 3), (1 / 3), -1'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '20'.

Long Form Response

11. **Answer:** 517. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 21,000 Cal. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key